TORQUE WRENCH CALIBRATION CERTIFICATE
교정성적서

DEMO CERTIFICATE — SIMULATED DATA ONLY

1. 기본정보 (General Information)

Certificate No.	DEMO-2026-0001	**Calibration Date**	2026-11-03
Customer	UNIVERSITY PROJECT	**Issue Date**	2026-11-03
Equipment	Torque Wrench	**Manufacturer**	DEMO TOOLS
Model	TW-100	**Serial Number**	TW2026-001
Maximum Capacity	100 N·m	**Calibration Range**	20 ~ 100 N·m
Direction	Clockwise (CW)	**Calibration Location**	Simulation Laboratory

2. 환경조건 (Environmental Conditions)

Temperature	23.0 °C	**Relative Humidity**	50 %RH

3. 교정 방법 (Calibration Method)

Method	Torque Wrench Calibration Simulation
Educational Reference	ISO 6789-2 (reference only)
Note	Simplified educational sequence with reference to ISO 6789-2. This simulation does not claim to fulfil all requirements of ISO 6789-2. 본 프로그램의 측정 시퀀스는 교육용으로 단순화된 절차입니다.

4. 교정 결과 (Calibration Results)

Nominal Torque (N·m)	Series 1 (N·m)	Series 2 (N·m)	Series 3 (N·m)	Mean (N·m)	Error (%)	U (k=2) (N·m)	Result
20.00	19.96	19.95	19.98	19.96	-0.18	0.05	**PASS**
40.00	40.05	40.11	40.07	40.08	+0.19	0.10	**PASS**
60.00	60.14	60.14	59.92	60.07	+0.11	0.20	**PASS**
80.00	79.72	80.01	79.82	79.85	-0.19	0.25	**PASS**
100.00	100.03	100.11	99.73	99.96	-0.04	0.32	**PASS**

Torque unit: N·m. Measured values are shown with two decimal places. Error (%) = (Mean − Nominal) / Nominal × 100. U is the expanded uncertainty (k = 2) in N·m.

5. 추가 결과 (Additional Results)

Calibration Direction	Clockwise (CW)
Number of Measurements	15 (3 series × 5 points)
Preloading Cycles	3 × 100 N·m (excluded from results)
Reference Instrument	Simulated Reference Torque Transducer (DEMO-REF-01)
Measurement Uncertainty	U = 0.24 % ~ 0.33 % of reading (k = 2, approx. 95 %); max 0.322 N·m
Tolerance	±4 % of nominal (educational assumption)
Conformity Assessment	**PASS - all points conform to the stated tolerance**
Decision Rule	Simple acceptance (단순 상대오차 판정): PASS if |relative error| ≤ tolerance. Uncertainty is reported but not taken into account.

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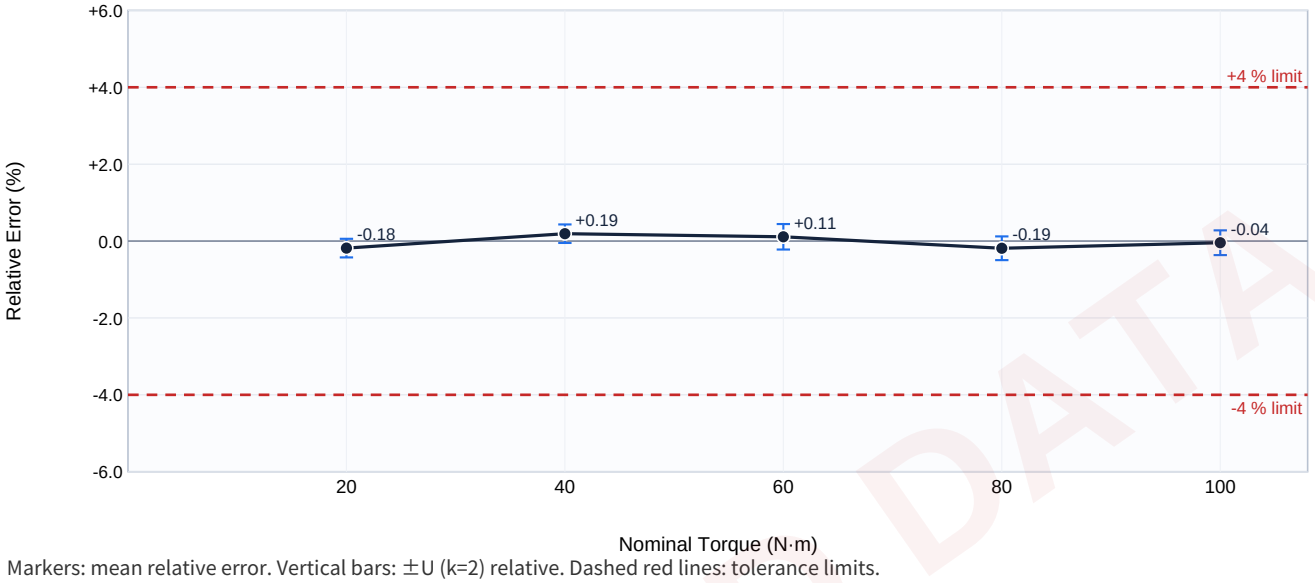
6. 상대오차 그래프 (Relative Error vs Nominal Torque)



Markers: mean relative error. Vertical bars: ±U (k=2) relative. Dashed red lines: tolerance limits.

7. 측정불확도 (Measurement Uncertainty Budget)

Nominal (N·m)	s (N·m)	uA (N·m)	uref (N·m)	ures (N·m)	uadd (N·m)	uc (N·m)	k	U (N·m)	U (%)
20.00	0.0153	0.0088	0.0200	0.0029	0.0100	0.0242	2	0.048	0.242
40.00	0.0306	0.0176	0.0400	0.0029	0.0200	0.0482	2	0.096	0.241
60.00	0.1270	0.0733	0.0600	0.0029	0.0300	0.0994	2	0.199	0.331
80.00	0.1473	0.0850	0.0800	0.0029	0.0400	0.1235	2	0.247	0.309
100.00	0.2003	0.1157	0.1000	0.0029	0.0500	0.1609	2	0.322	0.322

- Input quantities are assumed uncorrelated; sensitivity coefficients = 1.
- Type A: uA = s / √n (n = 3).
- Reference standard: U = 0.2 % of reading at k = 2 (educational assumption).
- Resolution r = 0.01 N·m, rectangular distribution, u = r / (2√3).
- Additional standard uncertainty = 0.05 % of reading (environment / reproducibility allowance, assumed).
- Expanded uncertainty U = k · uc with k = 2. With n = 3 the Type A component has only 2 degrees of freedom; a full GUM evaluation (Welch-Satterthwaite) would give a larger k. Simplified on purpose.

8. 비고 (Remarks)

1) All torque values in this document were produced by a software simulation (virtual torque measuring device). No physical instrument was connected.
2) The default tolerance (±4 %) is an educational assumption and is not a requirement that applies to every torque wrench. 기본 허용오차는 교육용 가정값입니다.
3) Preloading data are recorded separately and are not used for the mean value or the uncertainty evaluation.
4) Simulation record ID: 1; speed preset: Normal (timing only - does not affect results).

9. 서명 (Signatures)

Performed by	Student	**Reviewed by**	Instructor
Signature		**Signature**	
Date		**Date**	

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This document is generated for educational purposes and is not a valid calibration certificate.